

A 3-Sphere with One Imaginary Axis: Two Curvature Signs and an Equatorial S^2 , Not a Selection of de Sitter

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: [n4hy](#)

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Abstract

A guess: virtual modes live on a 3-sphere whose fourth embedding axis is imaginary and whose values are curvature, so the same sphere has de Sitter and anti-de Sitter regions. A guess is not a habitat and not a result. The geometry underneath it is real.

Round S^3 of radius ρ has Ricci scalar $6/\rho^2$, computed from the metric. $\rho \rightarrow i\ell$ sends $\mathcal{R} \rightarrow -6/\ell^2$. One family, two signs. The equator is a round S^2 . The Haar mean of the embedding coordinate $X_4 = \cos \chi$ is identically zero. All four identities **[P]**.

Identifying X_4 with curvature, and putting virtual particles on this S^3 , is a guess. It does not copy FGHMV onto de Sitter: $\dim \mathfrak{so}(1,4) = 10 < 15 = \dim \mathfrak{so}(2,4)$ is still Paper 17. It does not measure Λ .

1 The 3-geometry

$$ds^2 = \rho^2(d\chi^2 + \sin^2 \chi d\Omega_2^2).$$

Christoffel reduction of this metric gives $\mathcal{R} = 6/\rho^2$. The substitution $\rho = i\ell$ is algebraic and flips the sign. That is the spherical / hyperbolic pair: S^3 and H^3 . Calling the two signs dS and AdS *areas* is allowed as a name for the two 3-geometries. They are not yet 4d de Sitter and anti-de Sitter spacetimes.

The equator $\chi = \pi/2$ is S^2 . Papers 23–27 already work on a 2-sphere topology (the diamond waist). That is a kinship of topology, not a derivation of those lattice results from this embedding.

2 Mean zero

On the unit S^3 , $\langle \cos \chi \rangle = 0$ by oddness of the Haar integrand. If X_4 is curvature, the uniform mean is zero. That is the foam picture I want, *given* the identification. It is not a derivation of the identification, and it does not select $\Lambda > 0$. A mean-zero axis is Minkowski as the average, with both signs as fluctuations.

3 What this does not move

Paper 17 stands. Paper 19 still says the spinless Einstein+ $\Lambda > 0$ vacuum is de Sitter, as a metric identity. Paper 27 still says the CHM envelope beats a flat kernel on the *cut*, with weak tracking. This note records a guess and the identities it sits on. It does not finish the selection.

Admission. *I wrote down a guess. I derived the Wick sign and the Haar mean. I did not promote the guess to a habitat, and I did not write that de Sitter was selected.*